CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

November 2013

EE207 / EE-207.01 NETWORK ANALYSIS

Date: 21.11.2013, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q-1 (a) Name the four different types of dependent sources with their symbols in electric circuits. **01**

(b) (i) Obtain a single current source for the network shown in **figure I**. **01**

(ii) It is know that the potential difference of 12 Ω resistor in **figure II** is 48 V. Find the entering current I in **Figure.II** **01**

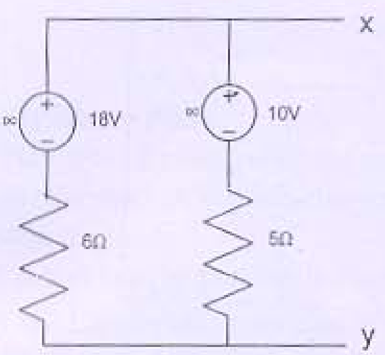

Figure I

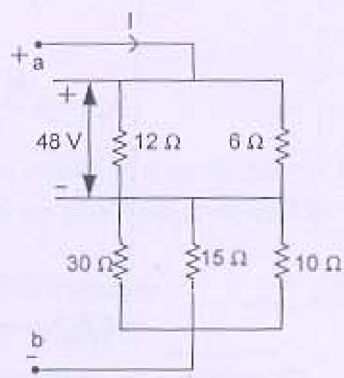

Figure II

(c) Explain with relevant diagrams: **04**

Voltage division and current division rule.

Q-2 (a) Find all the branch currents in the circuit of **figure III** using Nodal Analysis method. **07**

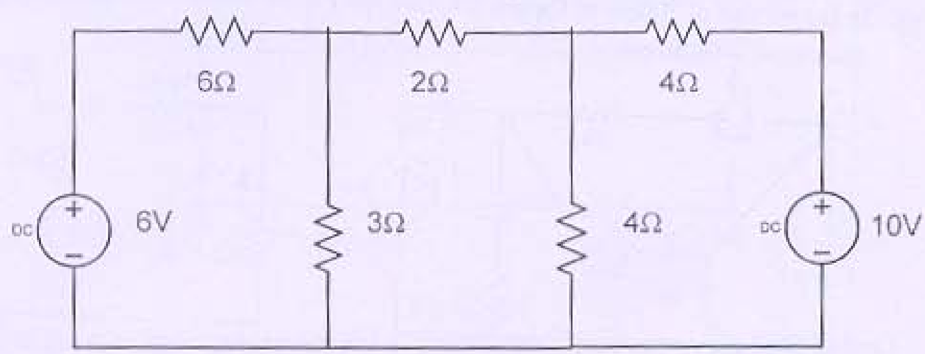

Figure III

(b) Write relation between Twigs and Links. Also write properties of a Tree in a Graph. **07**

OR

Q-2 (a) Determine the current in the 4Ω branch in the circuit shown in the **figure IV**. 07

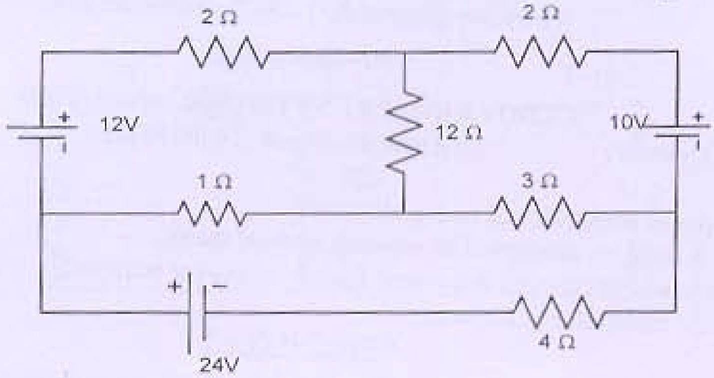

Figure IV

(b) **Figure V** represents a resistive network. Draw its graph. Select tree and obtain the tie-set matrix. 07

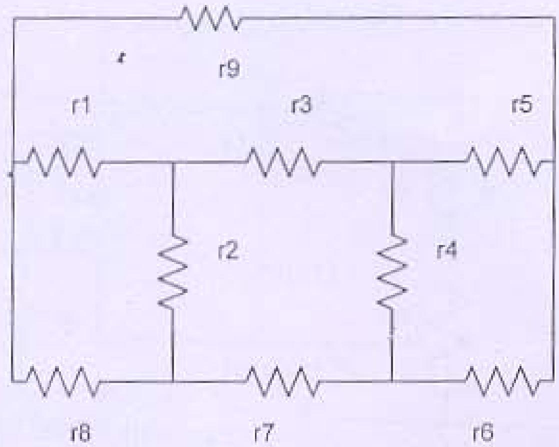

Figure V

Q-3 (a) Write statement of reciprocity theorem. Write the main steps for solving a network utilizing Reciprocity Theorem. 07

(b) In the circuit of figure of **figure VI**, find the power loss in the 1Ω resistor by Thevenin's Theorem. 07

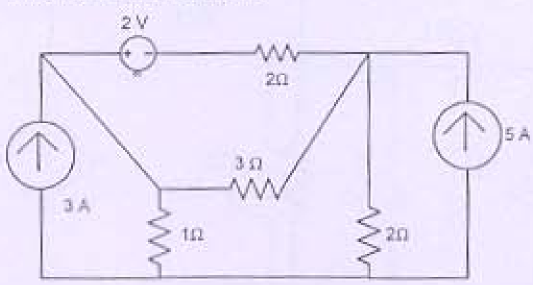

Figure VI

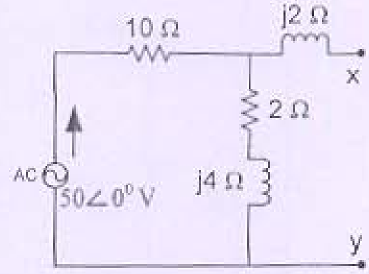

Figure VII

OR

Q-3 (a) Find the Norton's equivalent circuit across x-y terminal. Refer **figure VII**. 07

(b) Write statement of Maximum Power Transfer Theorem. Also prove the relevant equation of the Maximum power transfer theorem. 07

SECTION – II

Q-4(a) Define Gate function with its waveform. Synthesis the following waveform using Gate functions refer **Figure VIII**. 04

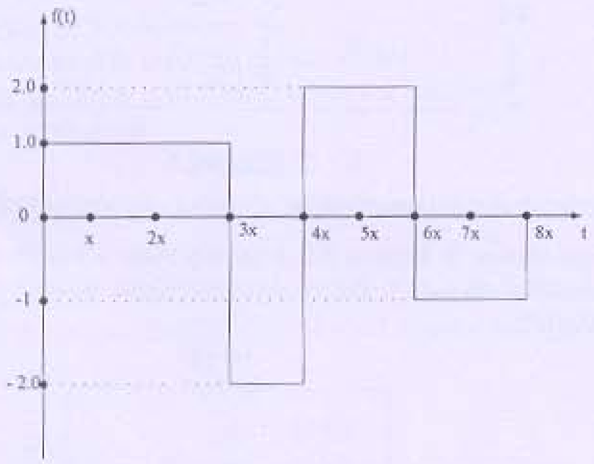

Figure VIII

(b) State True or False. 01

(a) For the h parameter being symmetrical the condition must be $\Delta h = -1$.

(b) For the ABCD parameter being reciprocal the condition must be $AD + BC = 1$.

(c) Fill in the blank. 02

i) A unit impulse function is obtained on the differentiation of __________ function.

ii) The Laplace transformation of $\cos\ (\omega_0 t + \phi)$ is ________.

Q-5 (a) Obtain the open circuit parameter from the network shown in **Figure IX** 07

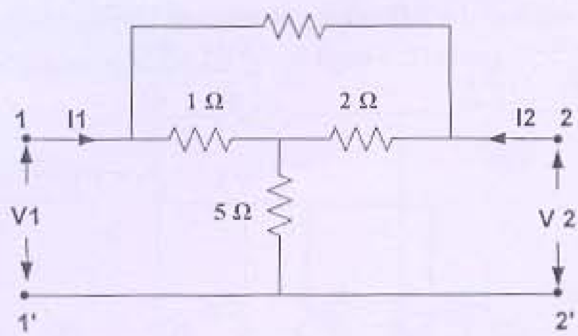

Figure IX

(b) Derive the relationship between H Parameter and Y parameter. (In both terms.) 07

OR

Q-5 (a) A Small signal equivalent circuit of CB mode of BJT is shown in **Figure X**. Assuming the voltage sources to be independent, find H parameters. 07

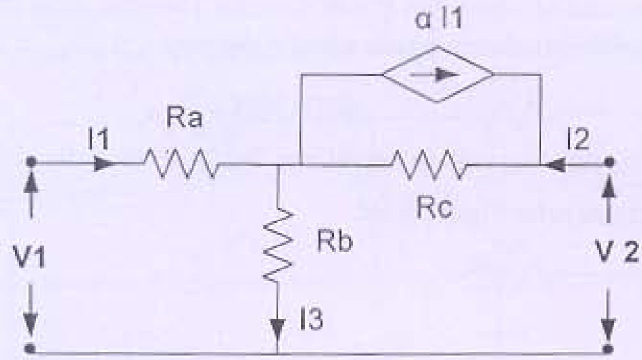

Figure X

(b) Classify the network configuration by giving its equivalent circuits. 07

Q-6 (a) In the network shown in **Figure XI.** A steady state is reached with switch K open. At t=0, the switch is closed. For the element values given , determine the values of $Va(0^-)$ and $Va(0^+)$. 07

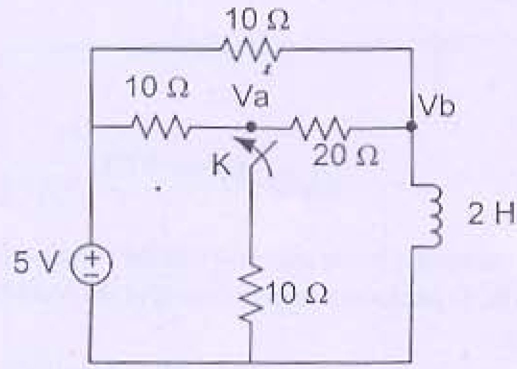

Figure XI

(b) Derive the equation of current flowing in RC series circuit while giving excitation as a step input. 07

OR

Q-6 (a) The network shown in **figure XII** has two independent node pairs. If the switch K is opened at $t = 0$,find the following quantities at $t = 0^+$, 07

(i) V_1, V_2 (ii) $\frac{dv_1}{dt}, \frac{dv_2}{dt}$

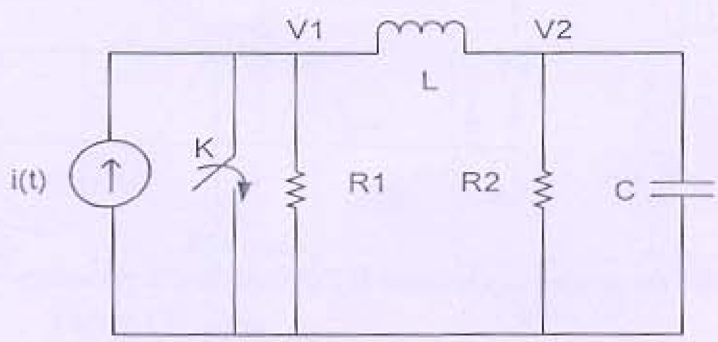

Figure XII

(b) Prove that $\lim_{s\to 0}[\, s\, F(s)] \;=\; \lim_{t\to\infty} f(t).$ 07